

Dear Hiring Manager,

I am writing to apply for the **Metrology Process Engineer** position at **Samsung Austin Semiconductor**. I hold a Master's degree in Chemistry and bring hands-on experience in **statistical data analysis, imaging-based metrology, calibration workflows, and yield improvement**, with a strong focus on translating measurement data into actionable process feedback in manufacturing-adjacent environments.

In my most recent role as a **Senior Research Engineer at Proteina Inc.**, I supported metrology and inspection workflows by developing **statistically grounded data pipelines** for optical and surface-based measurements. I worked extensively with **TIRF imaging systems**, using **Cr-grid reference materials** to perform **dimensional mapping, calibration verification, and field-uniformity analysis**, ensuring measurement accuracy and long-term data consistency.

I established **offline and inline characterization workflows** combining **optical microscopy, AFM, plate-reader assays, UV-Vis spectrometry, and ellipsometry** to enable fast inspection, measurement cross-validation, and root-cause analysis. By correlating dimensional maps and signal intensity distributions with process conditions, I helped identify **systematic measurement bias, drift, and equipment-induced variation**, directly supporting yield improvement and process stabilization.

A key part of my role involved **statistical interpretation of large measurement datasets**, including control charting, trend analysis, repeatability & reproducibility (R&R) assessment, and outlier diagnostics. I translated these results into **clear engineering reports and presentations** for R&D, manufacturing, and quality teams, enabling informed decisions on process adjustments, tool qualification, and disposition.

Earlier in my career at **Palogen LLC**, I supported semiconductor-adjacent reliability investigations, correlating **SEM/AFM imaging results with electrical test failures** and process parameters in collaboration with **SK Hynix and Amkor engineers**. This experience strengthened my ability to connect metrology signals with yield-impacting mechanisms under time-critical manufacturing conditions.

I am particularly drawn to this role because it sits at the intersection of **measurement science, statistical rigor, and manufacturing execution**. I am excited about the opportunity to contribute to Samsung Austin Semiconductor's commitment to world-class metrology, rapid issue detection, and continuous yield enhancement.

Thank you for your time and consideration. I would welcome the opportunity to further discuss how my background in **imaging-based metrology, calibration, and data-driven process improvement** can support SAS's fab operations.

Sincerely,
Dong Hyeun Lee